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MATH 014

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Equal Grades, Unequal Pay: Examining Internal Pay Disparities in the DC Government Workforce

Introduction

This is the DC Public Employee Salary Data set from Open Data DC, a DC government website operated by the Enterprise Data team from the Office of the Chief Technology Officer. This data is sourced from DC agencies and their partners to identify, curate, and distribute DC public data. This data set was originally published September 15, 2023, and was updated quarterly till September 18,2024, now its listed as updated as needed. It contains 40,6011 entries. It includes data from all DC agencies, but DC Public Schools (DCPS) and Metropolitan Police Department (MPD) make up the largest amount in entries. This Data set is useful for people interested in joining the DC Public employee sector by giving them a glimpse into their possible future salary. A dataset of this scale is also useful to examine how salary may range from position, agency, hiring date, etc.

Objective

The goal of this analysis is to evaluate this data set for any possible disparities to aid future job seekers. That way if they do employee they can properly evaluate how factors such as agency, hiring date, appointment type, and others may impact their salary. In times of a very tense job market, it's important to have these skills. My objective is to focus on hiring date, salary grade, and agency specifically. I felt as these variables would provide the most explanation. I believe that this can help me to understand and communicate how factors can impact salary on the broadest and most useful scope. I developed the following questions to help me do this "What is the correlation between hiring year and grade?" and "Do employees with different appointments earn systemically different rates in the same grade?". As my goal is to focus on the inequalities I believed that these two questions would allow me to sufficiently answer them.

Method

Data Understanding

The original Data set included 40,6011 rows and 9 columns, with 7 categorical and 2 numerical columns. Of the two numerical, comprate was a float data type and ObjectID was an integer data type. Most of the columns had at least 50 unique values which the exception of Government Type of Appointment (GVT_TYPE_OF_APPT). These value numbers ranged from 88 to 1,000.

The describe input revealed that the average salary across DC Public Employees in the data set was $89,087, when doing outside research I found that ABC7 reported that to live comfortably in DC a single person needs to be making at least $99,000. Their metric was the 50/30/20 rule which allocated 50% of income to necessities, 30% of income to discretionary spending, and 20% of income to savings or debt payments. This means that if we were to use this data to generalize DC public employees, the average DC public employees aren't making enough to live comfortably. However, based on the same article, they report that the median DC income is $67,992. This means that the average DC Public Employee is making more than the median income but not enough to necessarily live comfortably.

Data Cleaning

Detecting and Handling Missing Values

I found through using 'df.isnull().sum()' that the original data set was missing 40 values from the grade column. This column is essential for my analysis since across department and sector salary grade is used to organize salary data. Taking this into account, instead of dropping the column completely over this I opted to use the dropna().copy() to drop the missing values from the data set. This brough the new total of rows to 40,571.

Removing Duplicates

To find duplicates I used "df.duplicated().sum" and it found no duplicates, so I took no additional steps to remove them.

Column Conversions

I created a new column called GRADENUM from GRADE with all the most common grades due to frequency of unique values would make it difficult to map onto visualizations. I then dropped the GRADE column from the dataset. After that I converted GRADENUM into integer as I wanted to find summary statistics such as mean and median to use in my analysis, in this I removed all Grade Numbers that included letters and due to how little they were used in the dataset. They also usually were the same grades as other just for had the added letters to specify the department, but other columns such as DESCRSHORT already provided that information.

I also removed the OBJECTID column as it was a tally column for the rows in the dataset. This was redundant as Cocalc already listed the tally, and as I removed the rows missing a GRADE value it had an inaccurate count.

I converted the HIREDATE column into date time to allow me to use it for visualizations. I also isolated the year in it to create a new column that would allow me to use hiring year when creating visualizations.

While I created a COMPRATE_ROLLING column to help me analyze how year and month affected hiring due to the number of unique values I opted out of using it when creating visualizations. I mainly created that as the assignment requested temporal trends and seasonality detection.

Since I wanted to use job appointment type in my analysis, I created a column called main_types for the main appointments because after looking through each unique values outside of the four it

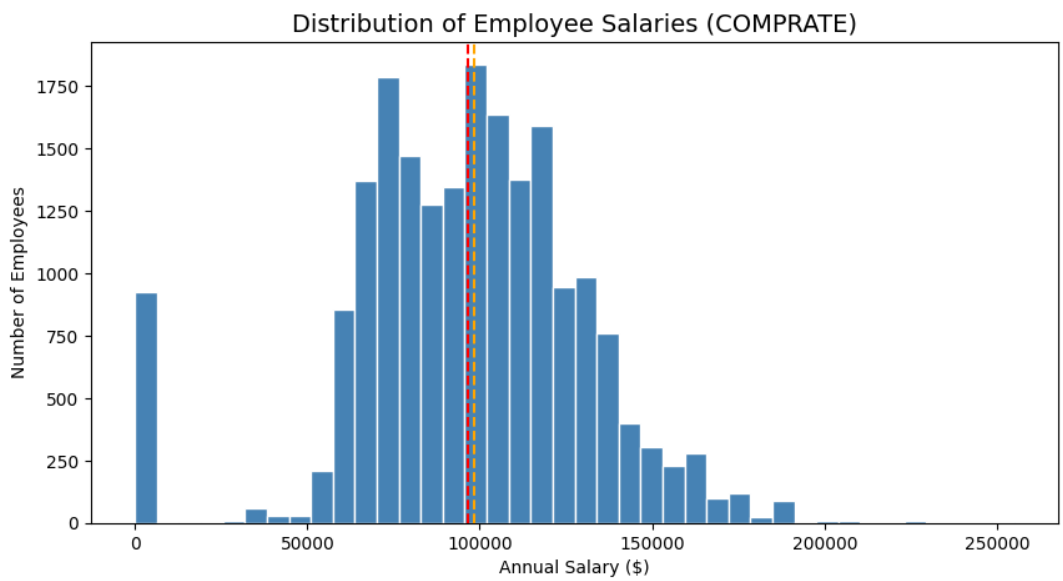
grabbed, the rest each represent less than 5% of the inputs in the dataset. This meant by creating this column I could still make summarizations based on it.

To standardize the JOBTITLE column, I opted to remove the white space and case sensitivity, because many jobs with the same title were coming up as different unique values.

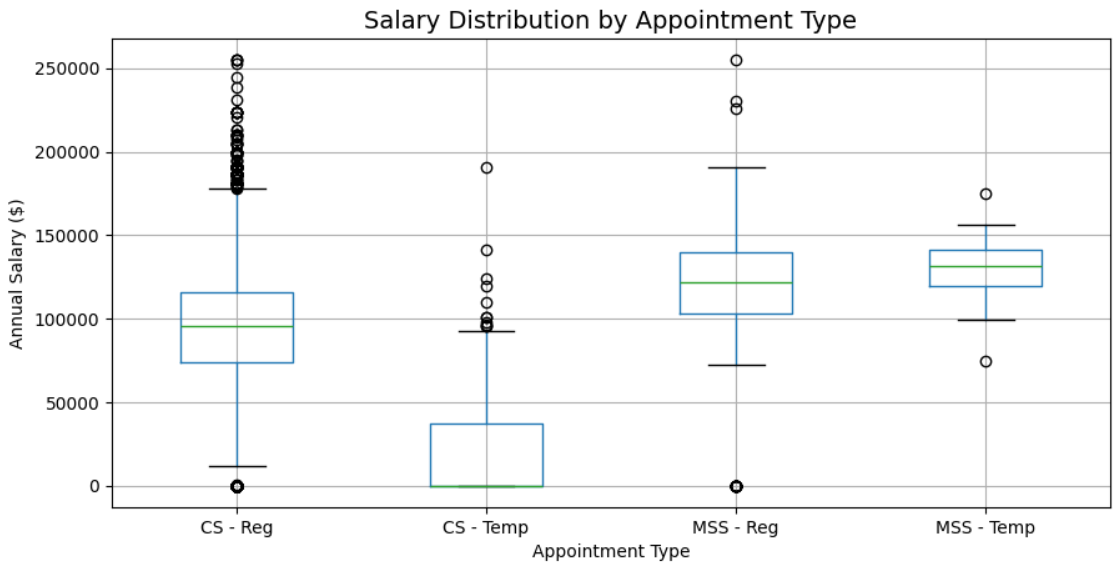
Any Processing Steps used

Prior to creating visualizations, I made a table grouping GRADE_NUM and COMPRATE by department using DESCRSHORT and I found the summary statistics which gave me a bigger picture of what the top ten departments were and what their average employee grade and salary were. To make sure that these were included in future visualizations, I created two new columns with top earners and bottom earners. I also used these columns to detect outliers and anomalies.

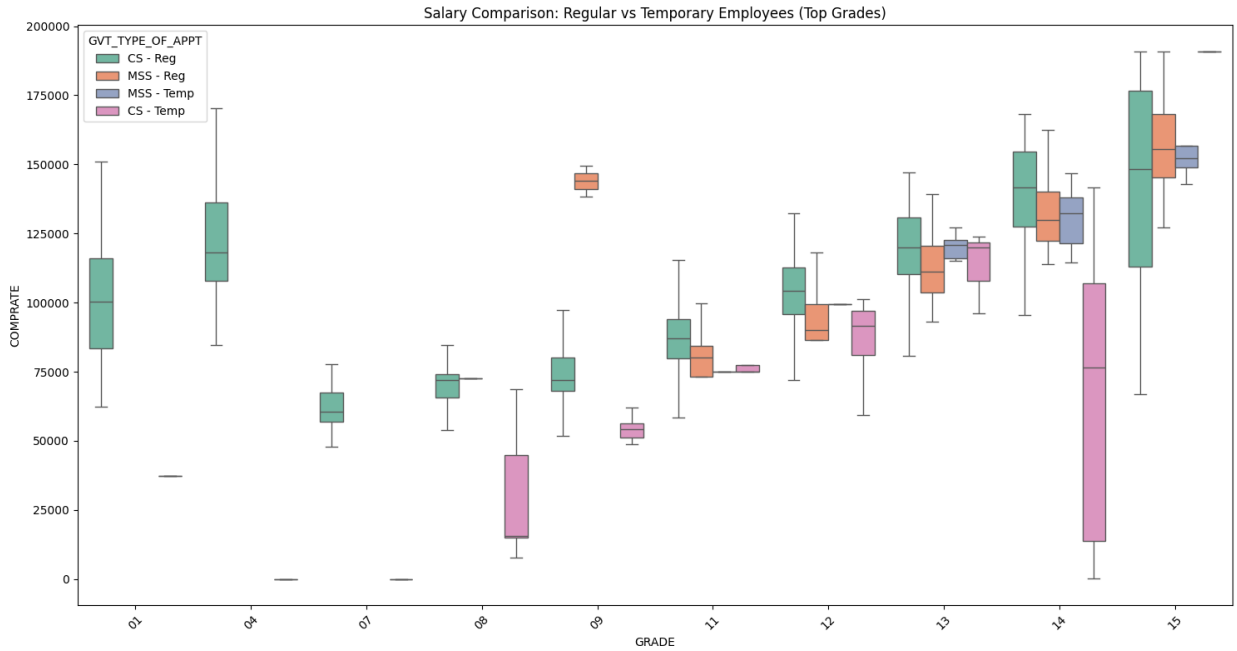
Key Visualizations



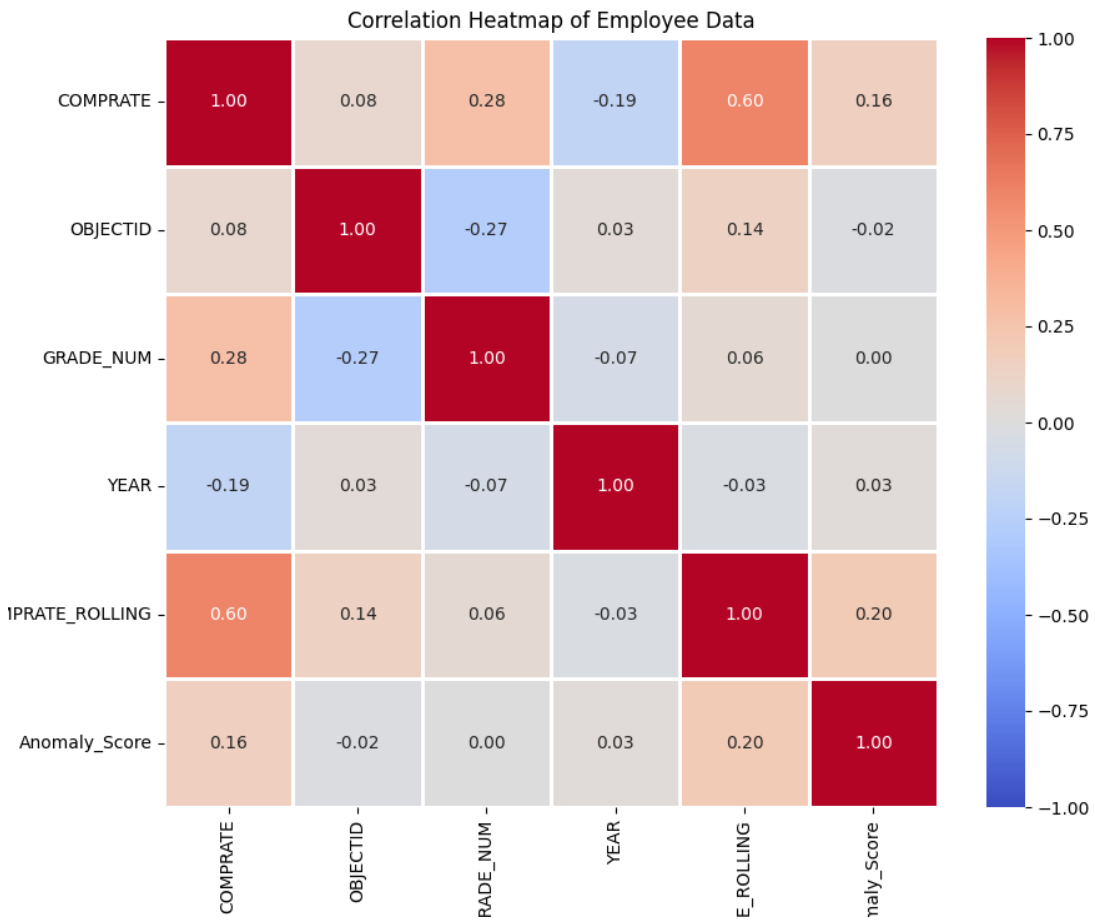
This univariable bar graph shows that the salary distribution is right-skewed, with the bulk of employees earning between $60,000 and $130,000. The mean ($89,087) exceeds the median, confirming the skew — a small number of very high earners pull the average upward. This suggests that central tendency statistics alone can be misleading when summarizing DC government pay. The biggest anomaly is how salaries from 0-50,000 and salaries above it are so isolated. This shows that around 1000 people's salaries



This bivariable box and whisker plot concludes that, CS - Reg and MSS - Reg employees have higher and tighter salary distributions, while temporary types show lower medians and wider spreads. MSS - Reg stands out with the highest median salary, consistent with its managerial nature. The presence of outliers across all types suggests that individual roles can command exceptional pay regardless of appointment category, but the systematic differences in medians confirm that appointment type is a meaningful predictor of compensation. The most significant outliers are on the CS-Reg value as it contains so many that they overlap. This shows that the people who exceed that plot can range very heavily.



This multivariable box and whisker plot shows that CS - Reg and MSS - Reg employees have higher and tighter salary distributions, while temporary types show lower medians and wider spreads. MSS - Reg stands out with the highest median salary, consistent with its managerial nature. The presence of outliers across all types suggests that individual roles can command exceptional pay regardless of appointment category, but the systematic differences in medians confirm that appointment type is a meaningful predictor of compensation. It directly answers my second question showing that as the grades go up CS in both types has more variation. The most apparent outliers are MSS-Reg for grade 9, CS-Reg for grade 11, MSS-Temp for grade 13, and MSS-Reg for grade 15 which all have a very narrow variation of salary rate.



The Red cells reveal that MSS - Reg employees consistently earn above their grade average — in some grades by tens of thousands of dollars. Blue cells show that temporary appointment types often earn below the grade mean. This confirms that the official grade structure does not fully determine pay: a second, informal pay hierarchy based on appointment type operates on top of it. For policymakers, this is the clearest evidence of systemic pay disparity within grade bands. This is an update from the heatmap included in my presentation to include more columns. Please disregard the inclusion of OBECTID, despite how its not present in my correlation matrix it still showed up on the graph.

Conclusions

Key findings are that appointment type out predicts hire year; the distribution is right-skewed; department creates a third axis of disparity; the appointment type pay hierarchy has been consistent over decades; and internal departmental variation is high. Highlights the surprise deviation heatmap as the most analytically powerful chart and explains what the violin plots revealed that simpler charts could not. COMPRATE excludes total compensation; the data is cross-sectional not longitudinal; grade comparability across agencies is assumed; filtering excluded some appointment types; and no demographic variables are available. There was little to no correlation between hiring year and salary grade.

With this conclusion, job seekers can assume that the biggest predictor of salary type and grade, and that when you're hired has no impact on your grade. It also shows that the biggest inequalities are visible in CS Temp as they have the lowest salaries. These positions still employ over 1000 employees though.

References

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